

NR4SD⁺

DP8 (8×4) — non-redundant signed digits of B with a carry, three per nibble

Step 1 — the dot product

$$\sum_{i=0}^7 A_i B_i$$

Step 2 — recode B into radix-4 digits and exchange the sums

Each B_i is a 4-bit value read as signed. Each pair plus its carry in gives a partial sum; a sum of 3 or more is pushed up as a carry, which leaves a digit in $\{-1, 0, 1, 2\}$. The pairs count the top bit as +8 and the signed read as -8: the difference -16 has the weight of the last carry, so together they are one extra digit:

$$s_i^j = 2B_i^{2j+1} + B_i^{2j} + c_i^j \quad c_i^{j+1} = \lceil s_i^j \rceil \quad d_i^j = s_i^j - 4c_i^{j+1} \quad c_i^0 = 0 \quad d_i^2 = c_i^2 - B_i^3 \in \{-1, 0, 1\}$$

$$\sum_{i=0}^7 A_i B_i = \sum_{i=0}^7 \left[\sum_{j=0}^2 4^j d_i^j \right] A_i = \sum_{j=0}^2 4^j \sum_{i=0}^7 d_i^j A_i$$

Two digits span $[-5, 10]$ and miss $-6, -7, -8$, so a signed nibble needs the third digit; when B_i is a lower slice of a wider B the top bit is read unsigned and $d_i^2 = c_i^2$.

Step 3 — the partial products

$$\sum_{i=0}^7 A_i B_i = \sum_{j=0}^1 4^j \sum_{i=0}^7 \underbrace{\left[d_i^j A_i \right]}_{R(B_i^{2j+1}, B_i^{2j}, c_i^j)} + 16 \sum_{i=0}^7 \underbrace{\left[d_i^2 A_i \right]}_{R_2(c_i^2, B_i^3)}$$

$R(B_i^{2j+1}, B_i^{2j}, c_i^j)$ depends on the pair and its carry in and takes one of four values; $R_2(c_i^2, B_i^3)$ is the top digit:

B_i^{2j+1}	B_i^{2j}	c_i^j	c_i^{j+1}	d_i^j	R		c_i^2	B_i^3	d_i^2	R_2
0	0	0	0	0	0					
0	0	1	0	+1	A_i					
0	1	0	0	+1	A_i		0	0	0	0
0	1	1	0	+2	$2A_i$		0	1	-1	$-A_i$
1	0	0	0	+2	$2A_i$		1	0	+1	A_i
1	0	1	1	-1	$-A_i$		1	1	0	0
1	1	0	1	-1	$-A_i$					
1	1	1	1	0	0					

Step 4 — split the work

The encoding is a function of B alone, so it is done in the dispatcher, which sends the digits d_i^j — the control signals of the selection multiplexers — to the PE; only the selection of the multiple stays inside:

$$\underbrace{\{B_i^{2j+1}, B_i^{2j}, c_i^j\} \longrightarrow d_i^j}_{\text{dispatcher: encoding}} \quad \underbrace{\{c_i^2, B_i^3\} \longrightarrow d_i^2}_{\text{PE: selection}} \longrightarrow \sum_{j=0}^2 4^j \sum_{i=0}^7 d_i^j A_i$$

Each d_i^j travels as a 2-bit code, $00 \rightarrow 0$, $01 \rightarrow +1$, $10 \rightarrow +2$, $11 \rightarrow -1$, which the cell uses directly: 6 control bits per lane. The digits depend on the carry, hence on every lower bit: a carry chain in the encoder, none in the cell.

B is the narrow operand: 3 digits × 8 lanes = 24 selections of 10 bits per DP8, each over $\{0, A_i, 2A_i, -A_i\}$ — the shift and the negation never combine, at the price of the third row. The two-digit form is the NR4SD⁺/Booth hybrid (nr4sd_2pp).